

RAGHUL ELAMATHI

INDUSTRIAL ENGINEER

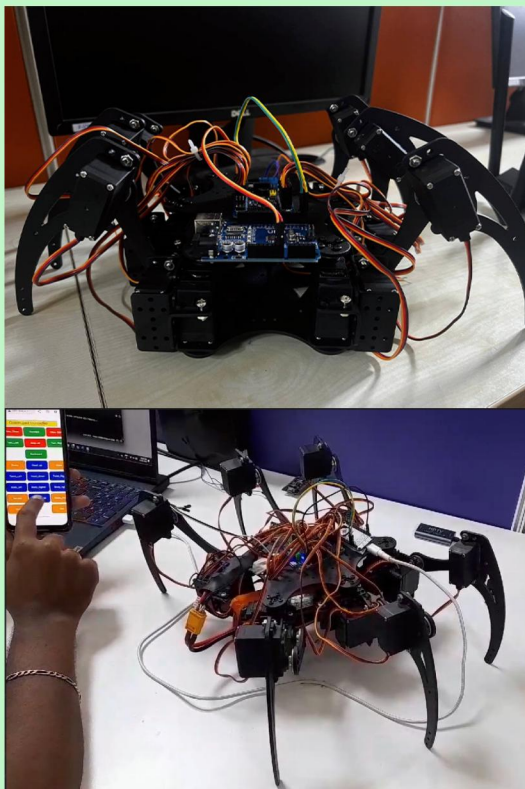
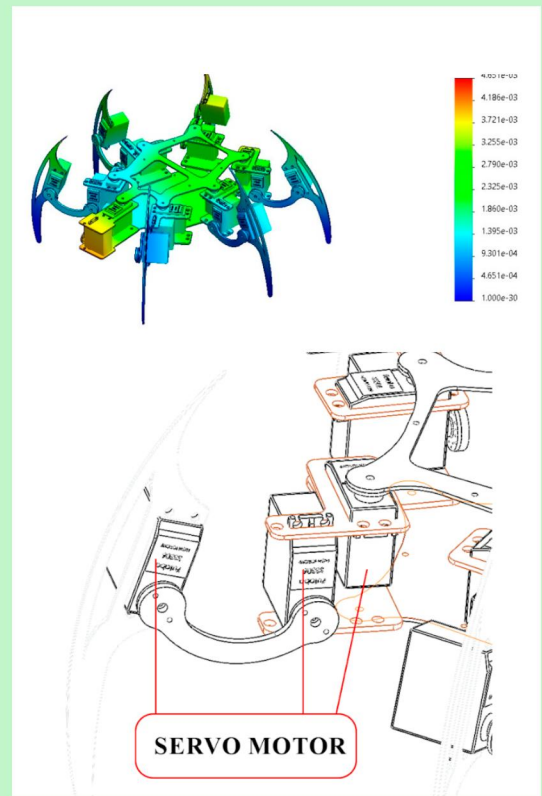
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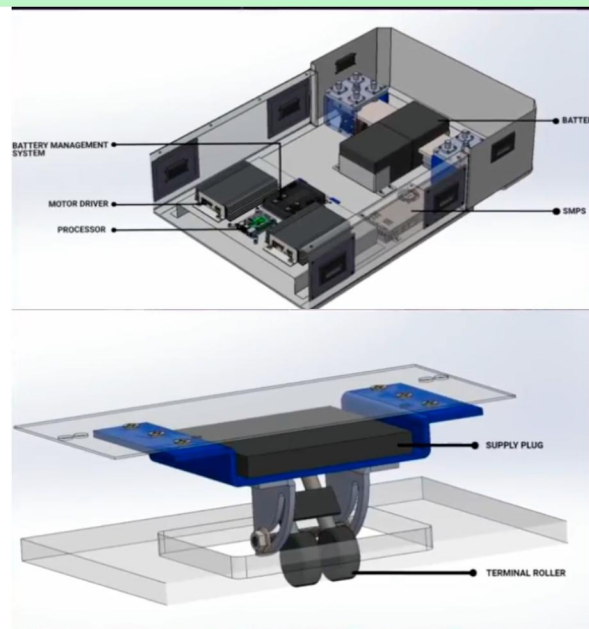
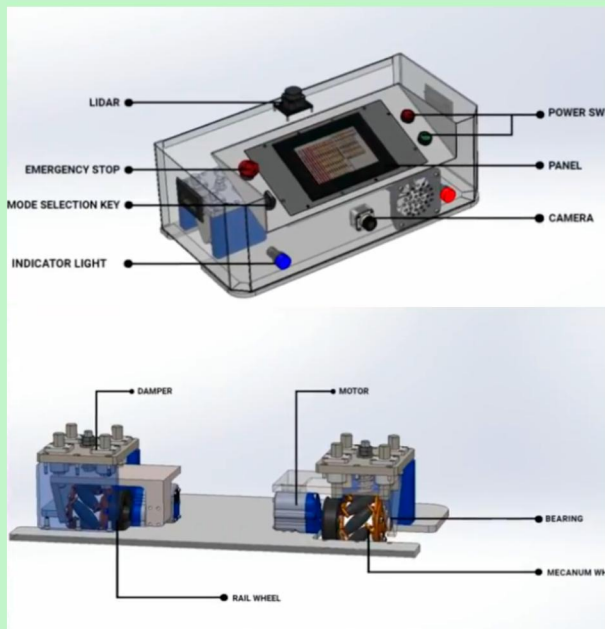
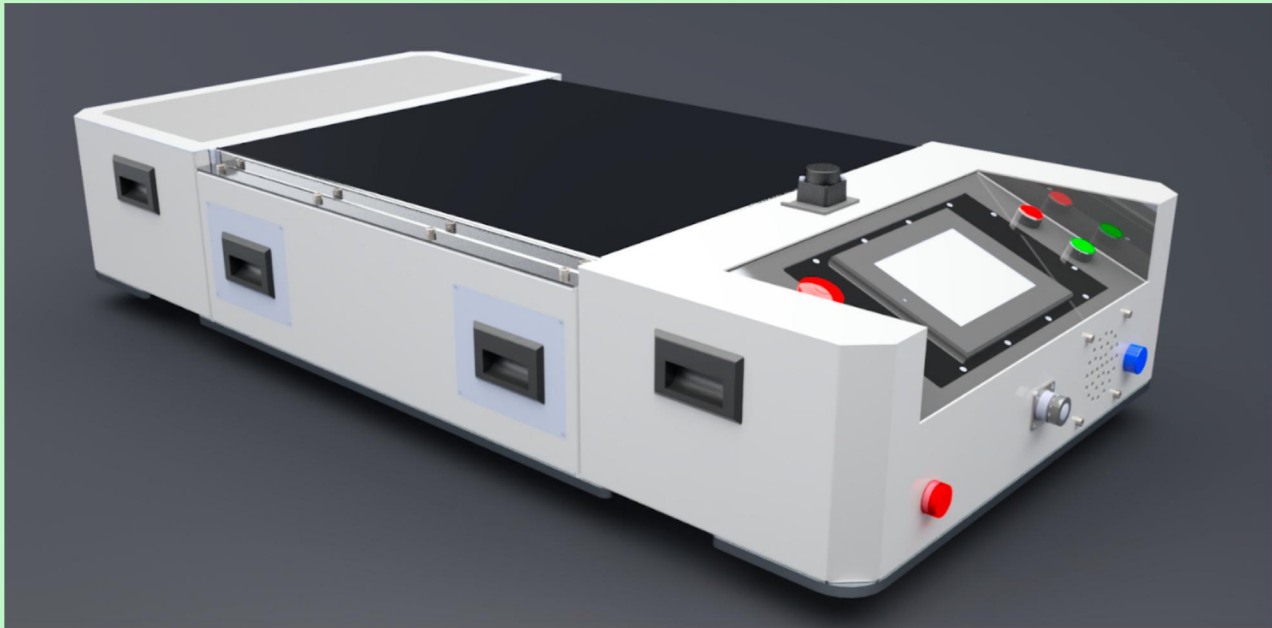
HEXABOT

Designed and developed a six-legged hexapod robot capable of stable locomotion on uneven terrain. The system was engineered to carry defense and delivery payloads, focusing on mechanical design, leg kinematics, load distribution, and CAD modeling for reliability and modular assembly.

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The hexapod was designed to operate in dual modes: fully autonomous and human-controlled. In autonomous mode, sensor inputs enable obstacle detection, gait planning, and stable navigation across complex terrains. In human control mode, the robot supports real-time teleoperation for precise maneuvering during critical tasks. The system integrates control algorithms, actuators, and communication modules, with extensive testing to validate stability, responsiveness, and operational reliability in practical scenarios.



AMR (AUTONOMOUS MOBILE ROBOT) 02.

Designed and developed an Autonomous Mobile Robot (AMR) for smart mobility applications, focusing on robust mechanical design, CAD modeling, and seamless sensor integration. The system enables autonomous navigation through real-time localization, path planning, and obstacle avoidance, ensuring safe and efficient movement in dynamic environments. A manual override mode allows controlled operation when required. The project emphasized system architecture, reliability, and performance optimization through iterative testing, making the AMR suitable for industrial and service-oriented applications.

SAE E-BAJA

Participated in SAE BAJA and Mega ATV competitions, including E-BAJA, where I worked on the design and development of off-road vehicles. Contributed to experimenting with an innovative 6-axis rear suspension concept to improve stability, traction, and terrain adaptability. The experience involved CAD design, basic CAE validation, fabrication support, and on-track testing, providing strong exposure to real-world vehicle dynamics, durability, and team-based engineering under competitive conditions.

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